

PROBABILISTIC APPROACH TO MACHINE LEARNING

What we have seen so far

- APPROACHING SUPERVISED LEARNING BY LINEAR MODELS
- EQUATION SOLVING LINEAR ALGEBRA APPROACH TO ML
- BASICS OF PROBABILITY, RANDOM VARIABLES, EXPECTATION
- METRICS, BIAS AND VARIANCE

Likelihood: Intuition

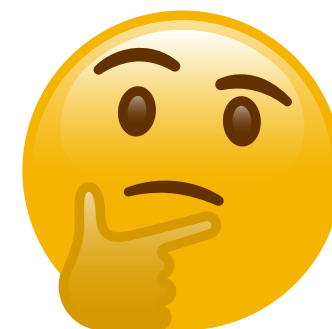


Samples A,B,C,... we
have with us



Probability distribution
with some parameters

**What is the probability that the sample A
came from the probability distribution we
thought of ?**



Likelihood: The Math

Let $X_1, X_2, \dots, X_n$ be i.i.d. samples from a distribution with PDF/PMF $f(x, \theta)$, where θ is the parameter of the distribution.

Since the samples are independent, the joint probability (density) of observing this particular dataset, as a function of the data, given a fixed θ , is

$$L(\theta \mid x_1, \dots, x_n) = P(X_1, \dots, X_n \mid \theta) = \prod_{i=1}^n f(x_i, \theta)$$

The likelihood function measures how plausible a specific parameter value is,
given a set of observed data

Maximum Likelihood Estimate: Intuition



Assuming the samples came from a distribution X , then what would have been the parameters of that distribution to give these samples ?

As the name suggests we try to find the parameters that maximise the likelihood of the samples for the parameters, as computed in the previous slides

Maximum Likelihood Estimate: The Math

Step 1: Compute the Likelihood

Given i.i.d. samples $x_1, x_2, \dots, x_n$ from a distribution with density $f(x | \theta)$, the likelihood is

$$L(\theta | x_1, \dots, x_n) = \prod_{i=1}^n f(x_i | \theta)$$

Step 2: Take the Log

Since log is monotonically increasing, maximizing $L(\theta)$ is equivalent to maximizing $\log L(\theta)$. The product becomes a sum:

$$\ell(\theta) = \log L(\theta | x_1, \dots, x_n) = \sum_{i=1}^n \log f(x_i | \theta)$$

Step 3: Differentiate w.r.t. θ and Solve

To find the maximizer, take the derivative of the log-likelihood with respect to θ , set it to zero, and solve:

$$\frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(x_i | \theta) = 0$$

Solving this equation for θ gives the maximum likelihood estimate:

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \ell(\theta)$$

Example Problem

A coin is tossed 5 times and the outcomes observed are H, T, H, H and T. We do not know if the coin is biased or not. Assuming probability of head being p , using the method of maximum likelihood estimation, estimate the value of p .

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TRY OUT

What would the estimate look like for another problem but with samples from Normal Distribution ?

Here is the solution for univariate Gaussian distribution
<https://statproofbook.github.io/P/ug-mle>

Application of MLE

Given the model of the form

$$y_i = f(\mathbf{x}_i; \mathbf{w}) + \epsilon_i = \mathbf{w}^T \mathbf{x}_i + \epsilon_i$$

$\mathbf{w}^T \mathbf{x}_i$: Deterministic Component

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

ϵ_i : Random Component

$$p(y_i | \mathbf{x}_i, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{w}^T \mathbf{x}_i, \sigma^2)$$

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Deterministic Component

$$\sigma^{2*} = \frac{1}{n} (\mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X} \mathbf{w}^*)$$

Noise Component

Mean and Covariance of w^*

$$\mathbf{E}\{w^*\} = w$$

w^* is an unbiased estimator of w

$$\text{cov}\{w^*\} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

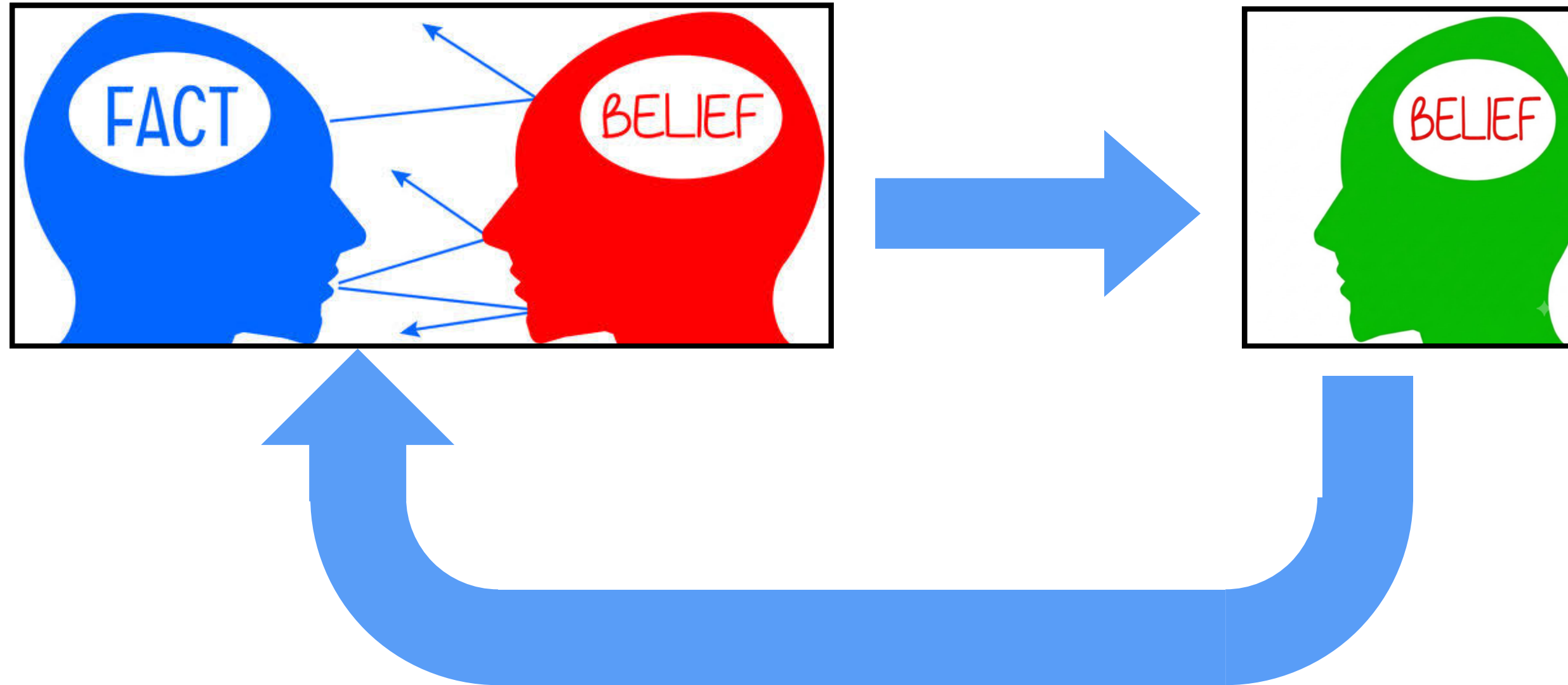
Since $\text{cov}(A, B) = \text{cov}(B, A)$
the covariance matrix is symmetric

Try to derive these expressions using the concepts taught previously in the tutorial

Problems with MLE

- **Ignores Prior Knowledge:** MLE only looks at current data and cannot integrate historical context, known physics, or expert beliefs.
- **Overfits on Small Data:** 3 heads in 3 flips makes MLE conclude tails is impossible ($p=1$)
- **MLE only provides point estimation of sample data.**

Bayes' Formula: Intuition



Old Belief = Prior
Fact/Evidence = Likelihood
New Belief = Posterior

Bayes' Formula: The Math

The diagram shows the Bayes' Formula equation with four labels and arrows pointing to specific parts of the formula:

- Prior**: An arrow points from this label to the $P(\theta)$ term in the numerator.
- Likelihood**: An arrow points from this label to the $P(Y|\theta)$ term in the numerator.
- Posterior**: An arrow points from this label to the $P(\theta|Y)$ term on the left side of the equation.
- Marginal Probability**: An arrow points from this label to the $P(Y)$ term in the denominator.

$$P(\theta|Y) = \frac{P(\theta) \cdot P(Y|\theta)}{P(Y)}$$

We can consider the denominator term to be a scaling factor and hence we can turn this equality into proportionality and thereby continue. Hence by doing this we ignore the marginal probability in the denominator.

Conjugate Priors

A prior distribution is said to be conjugate to a likelihood function if the resulting posterior belongs to the same distribution as that of prior

Prior	Likelihood
Gaussian	Gaussian
Beta	Binomial
Gamma	Gaussian
Dirichlet	Multinomial

References

<https://www.statlect.com/fundamentals-of-statistics/>

A First Course in Machine Learning by Simon Rogers

Course LMS Slides

APPENDIX

Bayesian Approach to ML: MAP

$$y = \mathbf{X}\mathbf{w} + \epsilon \quad \epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_N).$$

On solving using Bayes' formula we get

$$\Sigma_{\mathbf{w}} = \left(\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} + \Sigma_0^{-1} \right)^{-1}$$

$$\mu_{\mathbf{w}} = \Sigma_{\mathbf{w}} \left(\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{y} + \Sigma_0^{-1} \mu_0 \right)$$

$$p(\mathbf{y} \mid \mathbf{w}) = \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N)$$

Likelihood

$$p(\mathbf{w}) = \mathcal{N}(\mu_0, \Sigma_0),$$

Prior

Think about this

Till now everything we have seen -
MLE, Bayesian Approach does not
consider the order in which the
datapoints/ samples appear. Can you
think of some modification in our
formulation which can overcome this
flaw ?

